

Align The Misaligned

The internet has fragmented into distinct ecosystems across the globe. This article explores how ecosystem architecture can embrace this fragmentation as an opportunity for innovation rather than a problem to solve.

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It is remarkable how the history of virtual space mimics that of its physical counterpart. Like the universe's big bang, or better yet, the earth's tectonic shifts, our once one and only internet has shattered into many pieces. To lead with the bad news: no attempt to glue its pieces back together will ever mend it into a whole again. But the fact that something is broken does not mean that it is worthless. Changing our relation to fragments and waste is key to our planetary future. So why exclude the internet from our newfound consciousness?

This insight is key to the practice of ecosystem architecture: an approach that views digital landscapes as complex, interconnected systems akin to natural ecosystems. Just as natural ecosystems thrive on diversity and complex interdependencies, ecosystem

architecture posits that our digital ecosystems must embrace fragmentation and diversity as opportunities for growth and innovation rather than as obstacles to overcome.

While the existence and dangers of the Chinese and Russian internet are widely known and debated, the prevalence of this discourse obscures the fact that our 'normal' internet experience has been far from the norm globally for quite a while. In most Asian countries, so-called superapps have replaced browsers and apps as the primary way to interact with the web. Africa is pioneering the adoption of mesh and peer-to-peer networks that creatively give (some form of) internet access to people who cannot afford or rely on stable and persistent connections. Even our own mobile app store ecosystems should be understood as a break from the Open Global Web, which itself thus needs to be further predicated: let's call it the OG internet for now.

To understand how we arrived at this fragmented state, we need to look back at the internet's evolution, which has been far from uniform across the globe. Its history reflects both technological advancements and geopolitical realities, with roots tracing back to its very origins in the United States. While the ARPANET-DARPANET split in the United States represents an early instance of fragmentation, the global story is far more diverse and nuanced, with Europe developing unique academic networks, France pioneering Minitel, and Africa leapfrogging directly to mobile networks.

Even the debate around internet fragmentation is nothing new, dating back to the browser wars of the mid-1990s. However, it heated significantly around 2013, when Edward Snowden's revelations turned a gradual drift into a maelstrom of concerns about internet governance and privacy. To better understand the nature of this fragmentation, we can turn to a study from the World Economic Forum published in 2016. This research identified three distinct types of internet fragmentation:

Technical Fragmentation: Conditions in the underlying infrastructure that impede the ability of systems to fully interoperate and exchange data packets and of the Internet to function consistently at all endpoints.

Governmental Fragmentation: Government policies and actions that constrain or prevent certain uses of the Internet to create, distribute, or access information resources.

Commercial Fragmentation: Business practices that constrain or prevent certain uses of the Internet to create, distribute, or access information resources.

Recognizing the challenges posed by these forms of fragmentation, the Internet Society (ISOC) has put tremendous effort into making these concerns actionable and putting them on the political agenda. Its aim is to 'connect the unconnected' by fostering a more connected and interoperable online environment.

Despite these efforts, when mainstream media and public intellectuals assimilate the issue of internet fragmentation, a lot of the nuance and pragmatism gets lost in translation. More precisely, the status quo centers its narrative on restoring the OG

Internet. It focuses on what the web was rather than on what it can be. Proponents of this restorative stance can be roughly divided into three camps:

Past Dwellers: Pioneers who have a strong investment in, and are highly nostalgic for, the internet's early days. They consider the web's inception as an exemplary moment in history when free and open global discourse was, and could have stayed, a reality.

Power Mongers: Stakeholders whose (geo)political power can be challenged, stabilized, or expanded through control of the internet. These are mainly countries located in the Global North.

Gold Diggers: Parties that benefit economically from the OG internet. This camp mostly consists of companies that provide 'free' services that are paid for with data.

While these perspectives dominate much of the discourse around internet fragmentation, they often overlook the potential benefits and inevitability of some degree of fragmentation in a diverse global context. This brings us back to the importance of ecosystem architecture as a framework for understanding and navigating our current digital landscape.

While it's important to acknowledge the potential drawbacks of internet fragmentation, such as barriers to global communication or the creation of echo chambers, it's equally crucial to recognize the positive effects and innovative solutions that have emerged from this diversification. These include localized innovation, improved access in underserved areas, enhanced privacy and data sovereignty, cultural preservation, and potential resilience against cyber threats.

A critical reflection of the OG Internet's promises, the conditions under which this assumed historical exception could appear, and the reasons for its quick demise, is long overdue. As soon as the rest of the world got access to the web, the clash of civilizations immediately turned into a full-blown culture war. If anything, the internet's short history has made it crystal clear that open access to all information does not lead to universal consensus, trust, and understanding. Quite the opposite, it has led to an explosion of propaganda, fake news, and identity politics.

In light of these challenges and opportunities, the future of the internet, like any thriving ecosystem, lies in fostering connections that enhance mutual growth and understanding while respecting diversity. By adopting principles of ecosystem architecture, we can design a digital world that is more resilient, inclusive, and adaptive to the needs of all its inhabitants. This means not only acknowledging the fragmented reality but also leveraging it to build a more robust and dynamic internet for the future - one that celebrates innovation and diversity while addressing the challenges of fragmentation head-on.

